AI DRIVEN PRIVACY PRESERVING DIGITAL ECOSYSTEM FOR MODERNIZING SRI LANKA'S PADDY SUPPLY CHAIN

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Specialized in Software Engineering

Department of Software Engineering

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AI-DRIVEN PRIVACY-PRESERVING DIGITAL ECOSYSTEM FOR MODERNIZING SRI LANKA'S PADDY SUPPLY CHAIN

COMMON INTEGRATED SOLUTION REPORT

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Report submitted in partial fulfilment of the requirements for the

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Decisions along Sri Lanka's paddy supply chain, from crop protection and pricing to trading and national storage, are taken on information that is fragmented, unverifiable and rarely explained, while much of the information that would improve them is commercially or politically sensitive. This project presents Digital Goviya, an AI-driven, privacy-aware digital ecosystem of four independently deployable components integrated through role-based React Native mobile clients and a common HTTPS/REST interface. A digital farming component combines a ResNet50 disease classifier with an 85% confidence gate, an XGBoost yield regressor fed by ESP32 field telemetry with automatic weather-API fallback, and a rice variety recommender. A price forecasting component produces recursive multi-week district forecasts with XGBoost, explains them with SHAP and delivers evidence-constrained explanations in English and simplified Sinhala. A farmer-miller marketplace combines interpretable weighted matching, a federated-learning-style price reference, a retrieval-augmented assistant and validated two-agent negotiation. A warehouse coordination component anchors stock and redistribution records on a permissioned Hyperledger Fabric ledger, verifies with Groth16 zero-knowledge proofs that warehouse capacity meets a required threshold without disclosing the value, and scores warehouse reliability with a graph attention network. Evaluation recorded 90.91% disease classification accuracy, yield prediction with an R^2 of 0.989, sub-second responses for every measured interactive endpoint, zero false acceptances of invalid capacity claims, and detection of direct database tampering on a publicly deployed system. Learning models trained on synthetic data and the absence of completed field user studies limit claims about real-world impact. The project demonstrates that explainable, verifiable and privacy-scoped AI services can be composed into one coherent supply-chain ecosystem, and identifies validation on real data and verified cross-component data exchange as the next steps.

Keywords: paddy supply chain, explainable artificial intelligence, Internet of Things, permissioned blockchain, zero-knowledge proofs

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We are especially grateful to our external supervisor, Mr. Suranga Senanayake, whose knowledge of Paddy Marketing Board operations grounded our work in the realities of paddy procurement, storage and warehouse coordination, and whose observations directly informed the requirements of the warehouse coordination component.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
BCE	Binary Cross Entropy
CNN	Convolutional Neural Network
CORS	Cross-Origin Resource Sharing
EQ	Evaluation Question
FAISS	Facebook AI Similarity Search
FedAvg	Federated Averaging
GAT	Graph Attention Network
GNN	Graph Neural Network
GO	Group Objective
HTTPS	Hypertext Transfer Protocol Secure
IoT	Internet of Things
JSON	JavaScript Object Notation
JWT	JSON Web Token
LKR	Sri Lankan Rupee
LLM	Large Language Model
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NFR	Non-Functional Requirement
NPK	Nitrogen, Phosphorus and Potassium

Abbreviation	Meaning
PDPA	Personal Data Protection Act
PMB	Paddy Marketing Board
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
RMSE	Root Mean Squared Error
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SP	Subproblem
TLS	Transport Layer Security

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Rice is Sri Lanka's staple food, and paddy is the country's most widely cultivated crop. National paddy production reached approximately 5.05 million metric tons in 2025, with 2.74 million metric tons harvested in the 2025 Maha season and 2.31 million metric tons in the 2025 Yala season, and the 2025/26 Maha harvest was forecast at 2.82 million metric tons [1]. Behind these national figures sits a long chain of decisions taken by different actors: farmers decide what to plant, how to protect the crop and when to sell; rice millers decide whom to buy from and at what price; and the Paddy Marketing Board (PMB), which purchases paddy at a guaranteed price and releases stock to stabilise consumer prices, decides where stock is held and how it is moved when a warehouse is threatened [2].

Each of these decisions depends on information that is currently fragmented, delayed or unverifiable. Farmers diagnose leaf diseases by eye and estimate yields from experience, even as weather and soil conditions become more volatile. Price information reaches farmers informally and without any explanation of why prices are expected to move. Farmers and millers find each other through personal networks and telephone calls, and bargain without structured support. Within the PMB, warehouse supervisors report stock movements and available capacity by telephone and mobile messaging, so that redistribution decisions taken during floods or storage failures leave no independently auditable record [2]. Losses also accumulate after harvest: storage studies in the Anuradhapura District have documented both quantitative and qualitative losses of stored paddy under household conditions [3].

Digital agriculture research has shown that data-driven technologies can transform not only farm production but the relationships and governance of the whole food supply chain [4], and that machine learning, deep learning and the Internet of Things (IoT) can support crop monitoring, disease detection and yield estimation [5], [6], [7]. However, most existing work and most deployed tools address a single stage of the chain. Supply-chain reviews observe that research concentrates on traceability, provenance and production analytics, while downstream logistics coordination

receives comparatively little attention [8], [9]. Commercial farmer applications such as Plantix focus on crop diagnosis [10], and price forecasting, marketplaces and warehouse systems are typically built and evaluated in isolation.

This research project, titled "AI-Driven Privacy-Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain" and branded for users as **Digital Goviya**, responds to this fragmentation. It develops four complementary components, each researched and built by one member of the group, that together cover the paddy life cycle from the field to the national warehouse network: a digital farming advisory system (C02), an explainable paddy price forecasting service (C01), an AI-powered farmer-miller marketplace (C03) and a privacy-preserving warehouse coordination system (C04). The components share a common mobile presentation layer, a common REST and JSON integration style and a common commitment to explainable, verifiable and privacy-aware use of artificial intelligence (AI).

1.2 Stakeholders and user needs

The ecosystem serves three primary stakeholder groups and several secondary ones. Table 1.1 summarises the needs that each group brings to the system and the

component that primarily addresses them. The needs were identified from the four component studies, from consultation with the project's external supervisor, Mr. Suranga Senanayake, whose experience of PMB operations grounded the warehouse requirements, and from the literature reviewed in each individual thesis.

Stakeholder	Principal needs	Primary component(s)
Paddy farmers	Early and reliable disease diagnosis; yield expectations that reflect current field conditions; suitable rice varieties; understandable price outlook in Sinhala; easier access to buyers and fair negotiation	C02, C01, C03
Rice millers	Discovery of suppliers whose paddy type, quantity, district and price match purchasing requirements; structured negotiation without revealing private limits	C03
PMB Regional Managers	Verifiable stock reports; capacity claims that can be trusted during disasters; ranked redistribution options; defensible decision records	C04
PMB warehouse supervisors	Simple recording of stock events with supporting documents; protection of sensitive capacity figures from peer organisations	C04
Auditors and the Ministry of Agriculture	Read-only, tamper-evident audit trail independent of the application database	C04
Agricultural officers, collectors and traders (secondary)	District-level price trends with explanations; aggregated field and disease information	C01, C02

Table 1 : Stakeholders, confirmed needs and responsible components

Two needs cut across every group. First, users with limited technical background need outputs they can understand and act on, which requires bilingual presentation and explanations rather than bare numbers. Second, many of the data items involved (expected selling prices, negotiation limits, farm coordinates, warehouse capacities) are commercially or politically sensitive, so the system must reduce unnecessary disclosure while still allowing useful computation.

1.3 Alignment with the Sustainable Development Goals

The project is aligned with the United Nations 2030 Agenda for Sustainable Development [11]. Its intended contribution, stated here as a design intention rather than a demonstrated impact, is mapped to specific targets in Table 1.2.

SDG target	Relevance to paddy supply chain	Contributing components
2.3 Productivity and incomes of small-scale food producers	Better disease, yield and variety decisions; improved market access and price awareness for smallholders	C02, C01, C03
2.4 Sustainable and resilient food production	Targeted treatment after confident diagnosis instead of blanket chemical use; irrigation alerts from soil moisture	C02
2.c Functioning food commodity markets and timely market information, including on food reserves	Explained price forecasts; transparent farmer-miller matching; verifiable information on national paddy reserves	C01, C03, C04
12.3 Reduce food losses along production and supply chains, including post-harvest losses	Faster disease response in the field; evacuation of stock from flooded warehouses ranked by distance, capacity and reliability	C02, C04
9.c and 9.5 Access to ICT and upgraded technological capability	Mobile-first access to AI services in local languages; demonstration of advanced but low-cost cloud deployment	All

Table 2 : Alignment of the project with Sustainable Development Goal targets

Section 7.5 evaluates how plausible these contributions are, what could undermine them, and what evidence would be needed to claim them.

1.4 Purpose and scope of the common report

This common report presents the integrated problem, the four-component solution and the evidence for the solution as a whole. It explains why four components are needed, how they relate to one another, how they were integrated and evaluated, and what value the complete solution offers beyond its parts. It does not reproduce the detailed literature reviews, experiments and implementation descriptions of the components, which are presented in the four individual theses listed in Table 1.3. Readers are referred to those theses for component-level detail.

Team member	Student ID	Individual component and thesis
Ransara N.S	IT22102096	C01: AI-Driven Paddy Price Forecasting with Human-Readable Explanation
Kumarasingha P.A.N.D.	IT22294784	C02: IoT-Based Digital Dashboard for Paddy Farming (Digital Farming Advisory Guidance System)
Chamudi K.S.I.	IT22147950	C03: AI-Powered Farmer-Miller Marketplace with Privacy and Negotiation
Senarathne S. M. B. V. B.	IT22372444	C04: Privacy-Preserving Warehouse Coordination for National Paddy Distribution

Table 3 : Team members, components and individual theses

The remainder of the report is organised as follows. Chapter 2 defines the overall problem and its four subproblems, and states the group aim and objectives. Chapter 3 describes the overall solution and summarises each component. Chapter 4 explains how the components were integrated and how the system behaves as a whole. Chapter 5 describes the shared implementation and project execution. Chapter 6 presents integrated testing, evaluation and validation, including the achievement of the group objectives. Chapter 7 discusses commercialisation, sustainability and responsible innovation, and Chapter 8 concludes.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem addressed by this project is stated as follows:

> Decisions along Sri Lanka's paddy supply chain, from crop protection and harvest planning, through selling and buying, to national storage and redistribution, are taken on information that is fragmented across actors, often unverifiable, and rarely explained, while much of the information that would improve those decisions is commercially or politically sensitive and therefore withheld. There is no integrated, trustworthy digital decision layer that serves farmers, millers and the Paddy Marketing Board together.

The problem is deliberately stated independently of any technology. It concerns the quality, connectedness and trustworthiness of the information on which decisions are made, not the absence of a particular algorithm. The technologies adopted in Chapter 3 are evaluated by how far they improve these properties.

2.2 Root causes and overall research gap

Analysis across the four component studies identified five root causes that recur at every stage of the chain.

- **Isolated tools and data silos.** Farmers must use separate applications or none at all for weather, disease identification and yield estimation; price forecasts, marketplaces and warehouse records are held in unrelated systems. Outputs from one stage cannot inform the next.
- **Unexplained algorithmic outputs.** Where machine learning is used, it usually produces a single number (a disease class, a predicted price) without evidence that a non-technical user can interpret. Users therefore either ignore the output or over-trust it.
- **Fragile data acquisition.** Field IoT sensors lose power and connectivity, so systems that depend entirely on them fail exactly when conditions are worst.

- **Informal, unverifiable coordination.** Trading and warehouse coordination rely on telephone calls and personal relationships. Claims (a buyer's interest, a warehouse's capacity) cannot be checked, and decisions leave no auditable trail.
- **Disclosure as a precondition of coordination.** Existing systems generally require participants to reveal sensitive values (reservation prices, stock levels) to obtain the benefits of coordination, which discourages honest participation.

The four individual theses each identify a component-level gap: the lack of multimodal, fault-tolerant farm advisory systems (C02); the lack of explainable, multilingual paddy price forecasting for Sri Lanka (C01); the lack of an integrated marketplace that combines knowledge retrieval, price reference, matching and constrained negotiation (C03); and the absence of any system that combines a permissioned audit trail, zero-knowledge capacity verification and graph-based anomaly detection for warehouse coordination (C04). Taken together, these reveal an **overall research gap**: no existing work demonstrates an integrated ecosystem in which AI services at the farm, price, trade and storage stages are made explainable or verifiable, protect sensitive data by design, and are exposed through common interfaces so that they can serve the same users and feed one another. Solving any one stage in isolation would leave the decision chain broken at the others.

2.3 Subproblem and component mapping

The overall problem was decomposed into four non-overlapping subproblems, one per stage of the paddy life cycle, as shown in Figure 2.1. Each subproblem is owned by one component, while the shared concerns at the bottom of the figure are addressed jointly.

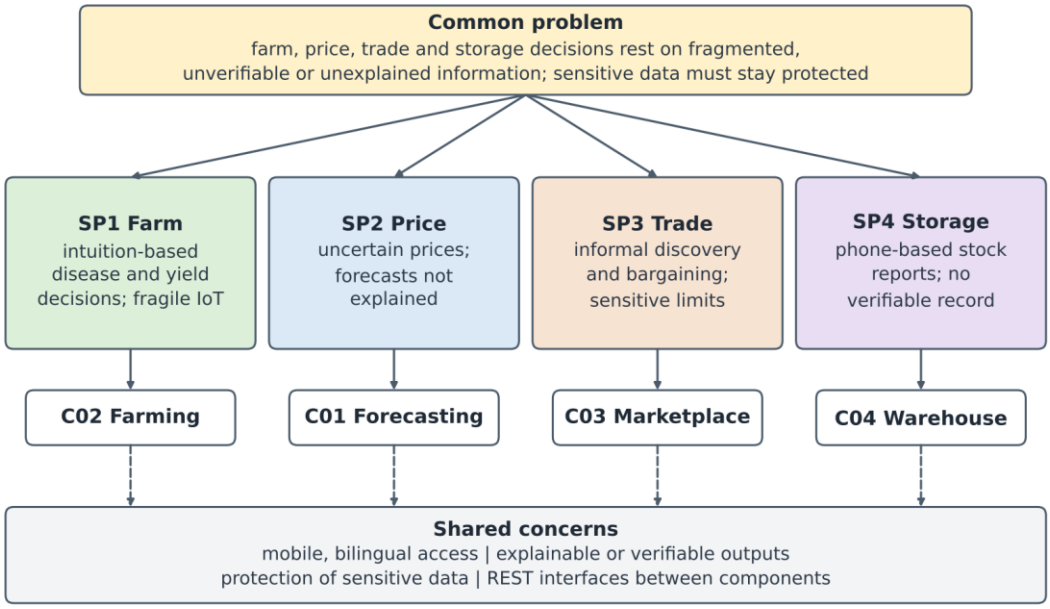


Figure 1 : Decomposition of the common problem into four subproblems and their components

- **SP1 Farm-level decision support (C02).** How can disease diagnosis, yield prediction and variety selection be combined with real-time field telemetry into one advisory service that continues working when field hardware fails?
- **SP2 Price uncertainty and explainability (C01).** How can district-level paddy prices be forecast several weeks ahead, and how can each forecast be explained, in English and simple Sinhala, without the explanation altering or inventing the prediction?
- **SP3 Market access and negotiation (C03).** How can farmers and millers discover compatible trading partners, obtain a price reference and market knowledge, and negotiate within private limits, in one marketplace workflow?
- **SP4 Verifiable and privacy-preserving storage coordination (C04).** How can warehouse stock events and redistribution decisions be recorded in a tamper-evident way, capacity claims be verified against a threshold without disclosure, and anomalous warehouse behaviour be detected from relational signals?

The subproblems are non-overlapping because each concerns a different decision, made by a different actor, at a different stage. They are interdependent because the output of each stage is the input context of the next: the expected yield bounds what a farmer can sell; the price outlook shapes the farmer's reserve price in negotiation;

traded and PMB-purchased volumes become warehouse inflows; and warehouse capacity and reliability determine how losses are contained after harvest. All four are necessary, because leaving any stage unaddressed would leave users to bridge that gap manually with the same informal, unverifiable information the project aims to replace.

2.4 Overall aim and group objectives

The common aim of the project is:

> *To design, implement and evaluate an integrated, AI-driven and privacy-aware digital ecosystem, Digital Goviya, that supports better-informed and more trustworthy decisions by farmers, millers and the Paddy Marketing Board across the farming, pricing, trading and storage stages of Sri Lanka's paddy supply chain.*

The aim is pursued through six measurable group objectives. Table 2.1 maps each objective to the components that deliver it and to the evidence used to evaluate it in Chapter 6.

ID	Group objective	Components	Evaluation evidence
GO1	Deliver AI-based decision support at each of the four supply-chain stages, each evaluated with metrics appropriate to its task	C01, C02, C03, C04	Classification and regression metrics, matching and negotiation rules, proof and integrity tests (Section 6.3)
GO2	Make every AI output explainable or verifiable to its user, and avoid presenting estimates as guarantees	C01, C02, C03, C04	SHAP and XAI outputs, confidence gate, rule-based match scores, validated negotiation, proof and digest checks
GO3	Protect sensitive data by design: negotiation limits, farm data, capacity values and personal data	C03, C04, C02	Limit validation and non-disclosure, ledger contents audit, transport security, role-based access tests
GO4	Provide accessible mobile, role-based and bilingual (English and Sinhala) user access	All	Implemented interfaces; language support; role access tests
GO5	Expose every component through documented, validated REST/JSON interfaces so that components can be composed into one ecosystem	All	Integration tests per component; interface specification (Appendix A); deployment evidence
GO6	Achieve responsive and fault-tolerant operation suitable for rural mobile use	All	API latency and repeated-request tests; sensor fallback test; container recovery

Table 4 : Group objectives, contributing components and evaluation evidence

2.5 Scope assumptions and constraints

In scope. The project covers paddy (not other crops) in Sri Lanka; the four decision stages described above; the research, design, implementation and technical evaluation of the four components; their exposure through REST interfaces and mobile clients; and a public cloud deployment of the warehouse component as a demonstration of the full stack under production-like conditions.

Out of scope. Payments and logistics execution between farmers and millers; integration with official government information systems; real-time crop pest detection beyond the three trained disease classes; and any claim that the system increases farmer income or guarantees fair prices, which would require longitudinal field evidence.

Assumptions. Farmers and millers have access to a smartphone with 3G or 4G connectivity; users enter reasonably accurate listing and field information; the historical price data used by C01 is representative of district price behaviour; and the synthetic data used where real data was unavailable (C02 environmental features, C03 price reference, C04 warehouse graph) is adequate to demonstrate the mechanisms, although not to establish real-world performance.

Constraints. The project was conducted within one academic year by four students with no budget for commercial infrastructure, which drove the choice of open-source frameworks and free-tier cloud hosting. Real PMB transaction data, real farmer-miller transaction data and multi-season field trials were not available, which is the single most important constraint on the evaluation. External AI services (OpenAI and other language model providers, weather APIs) introduce dependencies on availability, cost and credentials.

Dependencies. C02 depends on the Firebase Realtime Database and an external weather API; C01 and C03 depend on language-model services for explanations and assistance; C04 depends on a running Hyperledger Fabric network; and all components depend on the shared mobile client and gateway conventions described in Chapter 4.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

Digital Goviya is designed as one ecosystem with four specialised services rather than as four unrelated products. Its central value proposition is that every significant decision along the paddy supply chain can be supported by an AI output that is either **explained** (the user is shown why) or **verified** (the system can prove that a claim or record is genuine), delivered through a mobile interface in the user's language, without requiring participants to disclose more sensitive information than necessary.

The end-to-end outcome is a continuous decision path for the same crop. A farmer is advised on disease, yield and variety during the season (C02); consults an explained price outlook before selling (C01); lists the harvest, finds and negotiates with a compatible miller (C03); and the PMB's purchases and stock movements are recorded in a tamper-evident ledger that also supports verifiable emergency redistribution (C04). Each component can be deployed and used independently, which reduces integration risk and allows incremental adoption, but each also exposes outputs that the others can consume.

Four design principles were agreed by the group and applied to all components:

- **Evidence with every output.** Predictions are returned together with their supporting evidence: a confidence value and gate (C02), SHAP feature contributions and trend, risk and confidence categories (C01), a rule-based compatibility score and validated negotiation steps (C03), and proof verification results and integrity digests (C04).
- **Estimates are not guarantees.** Forecasts, price references and reliability scores are presented as decision support. An agent-generated agreement is not treated as a completed sale, and a low reliability score is not treated as proof of misconduct.
- **Minimum necessary disclosure.** Negotiation limits stay on the owner's side of the validation layer; capacity values never reach the shared ledger; personal data is never written on chain.

- **Independent services, common contracts.** Each component is a separately deployable service with its own data store, exposing schema-validated JSON over HTTPS.

3.2 High level architecture

Figure 3.1 shows the high-level architecture. It has five tiers: users and stakeholders; a presentation tier of role-based React Native (Expo) mobile clients; an HTTPS/REST boundary through which all client traffic passes; an application tier containing the four component services; and a data and trust tier in which each component owns its own store. Edge devices and external services are shown at the bottom.

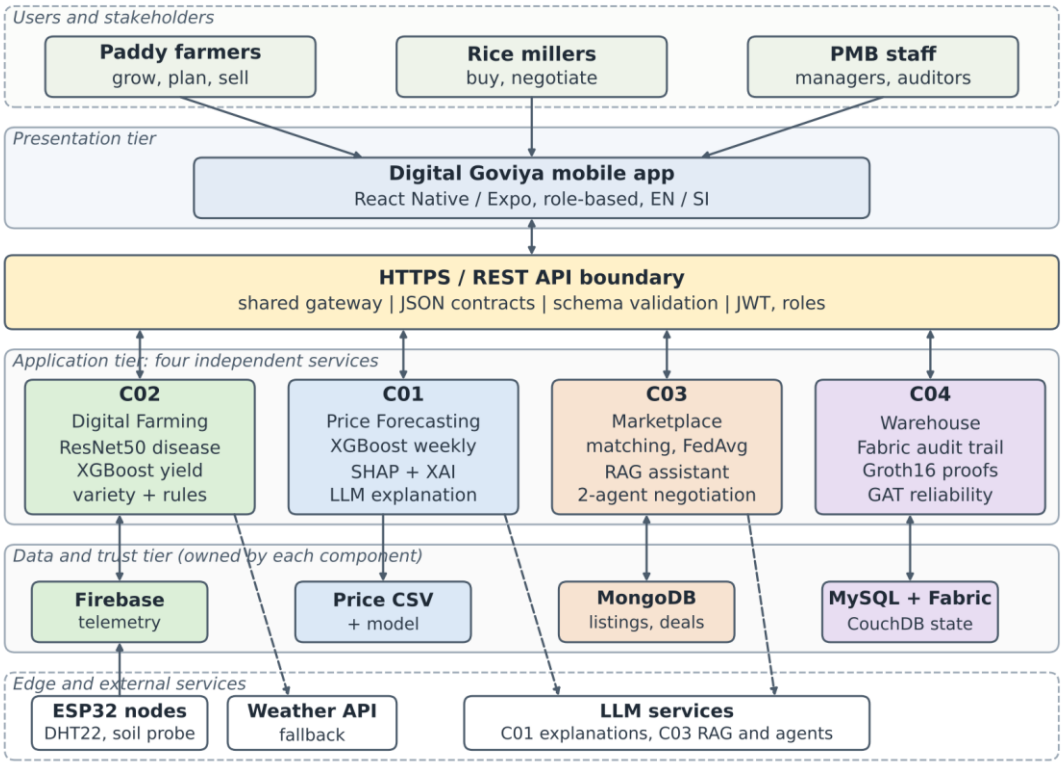


Figure 2 : High-level architecture of the Digital Goviya ecosystem

The architecture deliberately avoids a single shared database. Each component's data has a different shape and a different trust requirement: field telemetry is a high-frequency stream (Firebase Realtime Database), price history is a curated time series with a serialised model (CSV and model artefact), marketplace records are flexible documents (MongoDB), and warehouse records must be both relationally queryable and independently tamper-evident (MySQL anchored to a Hyperledger Fabric ledger

[12]). Integration is therefore achieved at the API boundary rather than at the database, which is the pattern recommended for combining off-chain AI inference with on-chain anchoring [13] and which keeps each component's security assumptions local.

The deployment boundaries follow the services. The C04 stack, comprising a TLS-terminating reverse proxy, the Node.js backend, the Graph Attention Network service, MySQL and a two-organisation Fabric network, is deployed as nine Docker containers on an Oracle Cloud Infrastructure instance served over HTTPS. The C01 and C02 Python services and the C03 Express and FastAPI services run as independent API processes that the mobile clients address through configured base URLs. Section 5.3 describes the deployment configuration in detail.

3.3 Component 1 summary : Paddy Price Forecasting (C01)

Owner. Ransara N.S (IT22102096).

Subproblem. SP2, price uncertainty and the lack of explainable, accessible forecasts.

Input. A district, a start date and a forecast horizon in weeks; historical weekly paddy price data for supported districts stored as a cleaned CSV dataset.

Core method. A trained XGBoost regression model [14] predicts the average paddy price for a district. A recursive multi-week mechanism feeds each weekly prediction back into the feature sequence to produce the next week's estimate. SHAP (SHapley Additive exPlanations) [15] attributes each prediction to its input features, and dedicated explainable-AI (XAI) modules convert this evidence into a trend class, an application-level confidence category, a risk level, a market outlook and recommendations. Finally, a large language model (LLM) explanation layer turns the structured evidence into short explanations in English and simplified Sinhala, under the constraint that it communicates the evidence and does not change or invent the prediction.

Output. JSON responses from four endpoints: `/predict`, `/forecast`, `/explanation/evidence` and `/explanation/llm`.

Contribution to the whole. C01 supplies the forward-looking market signal of the ecosystem. Its forecasts and explanations give farmers a reasoned expectation of price

movement before they enter the marketplace, and give warehouse and trading stakeholders a view of likely price trends. Figure 3.2 shows the component's layered architecture.

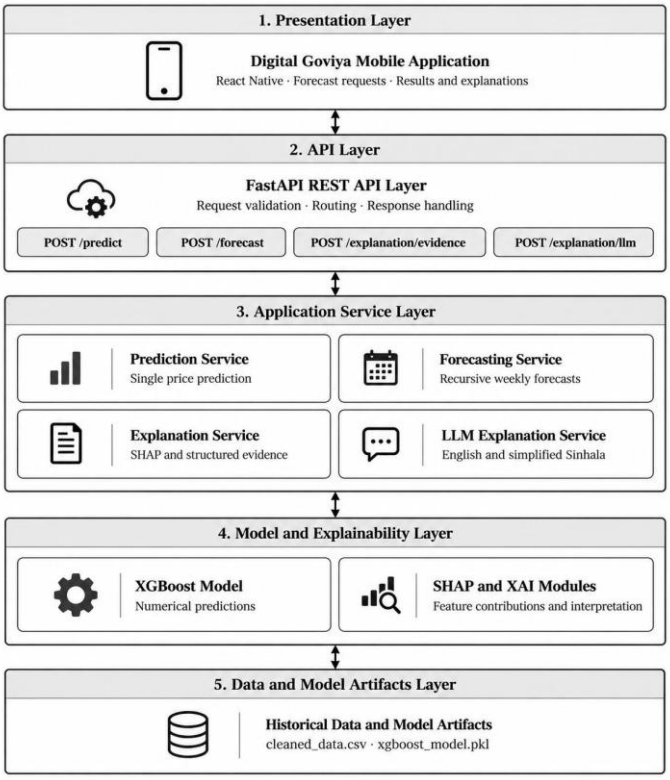


Figure 3 : Layered architecture of the C01 price forecasting and explanation service

3.4 Component 2 summary : Digital Farming (C02)

Owner. Kumarasingha P.A.N.D. (IT22294784).

Subproblem. SP1, farm-level decisions that rely on intuition and fragile, siloed tools.

Input. Smartphone photographs of paddy leaves; farm location and context (district, season, zone, latitude and longitude, soil salinity and iron toxicity); and live telemetry of temperature, humidity and soil moisture from ESP32 field nodes with DHT22 and capacitive soil-moisture sensors, streamed to the Firebase Realtime Database.

Core method. A ResNet50 convolutional neural network [16] fine-tuned by transfer learning from ImageNet classifies leaf diseases and withholds any diagnosis whose highest softmax probability is below 0.85, returning "Unknown Type" instead. An XGBoost regressor predicts yield in kilograms per hectare from scaled geographic and live environmental features. A scikit-learn pipeline with one-hot encoding and a

Random Forest classifier [17] recommends a rice variety and enriches it with growth duration and flood tolerance from a variety catalogue. A rule engine turns combined outputs into actionable alerts, for example an irrigation warning when soil moisture falls below 20% and yield is predicted to fall. If Firebase returns no telemetry because a field node has failed, the FastAPI backend automatically substitutes weather data from an external weather API so that predictions continue.

Output. JSON advisory payloads from endpoints including `/predict_disease` and `/predict_yield_production`, rendered as dashboards and alerts.

Contribution to the whole. C02 is the production end of the ecosystem. It protects the crop, estimates how much paddy will reach the market, and provides the field data foundation that later stages build upon. Figure 3.3 shows the assembled IoT prototype.

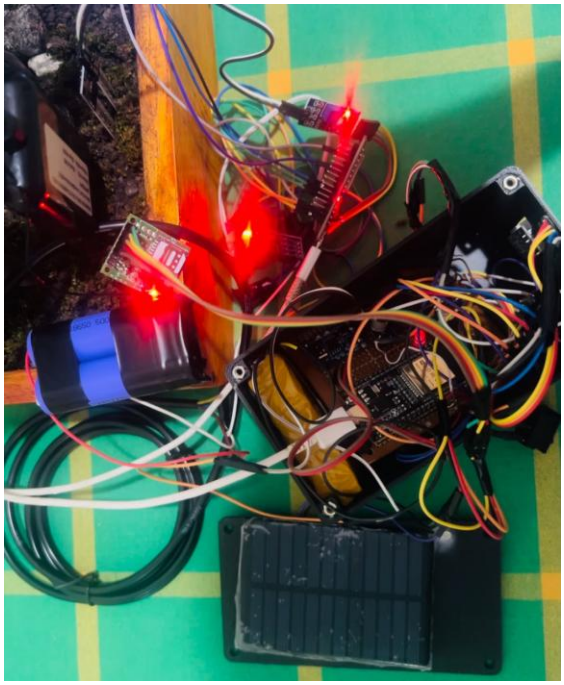


Figure 4 : Assembled ESP32 IoT field-node prototype used by C02 (original in colour)

3.5 Component 3 summary : Farmer-Miller Marketplace (C03)

Owner. Chamudi K.S.I. (IT22147950).

Subproblem. SP3, informal partner discovery and unstructured bargaining in which private limits are commercially sensitive.

Input. Farmer harvest listings (paddy type, quantity, district, expected price) and miller demand listings; user questions in English, Sinhala or Singlish.

Core method. A weighted, interpretable matching algorithm scores harvest-demand pairs out of 100 (district 40, paddy type 30, price closeness 20, quantity compatibility 10) and classifies them as Highly Recommended (85 and above), Recommended (65 to 84) or Moderate. A price-reference model trained by a simplified Federated Averaging (FedAvg) simulation [18] across four simulated district clients provides an indicative price at listing time. A retrieval-augmented generation (RAG) assistant [19] retrieves from a curated knowledge base of 32 records using multilingual embeddings and a FAISS index before generating an answer with GPT-4.1-mini. A two-agent negotiation service in FastAPI, in which farmer and miller agents are driven by a locally hosted LLM through Ollama, proposes, counters, accepts or rejects offers over at most six rounds, while a deterministic validation layer enforces each party's private minimum or maximum price [20].

Output. Ranked matches, price references, grounded answers and negotiation outcomes delivered through an Expo mobile application backed by an Express API and MongoDB.

Contribution to the whole. C03 is the market interface of the ecosystem, where farm output (C02) and price expectations (C01) are converted into trade, and where traded volume enters the storage stage (C04). Figure 3.4 shows its transaction life cycle, which deliberately separates an agreement from a completed sale.

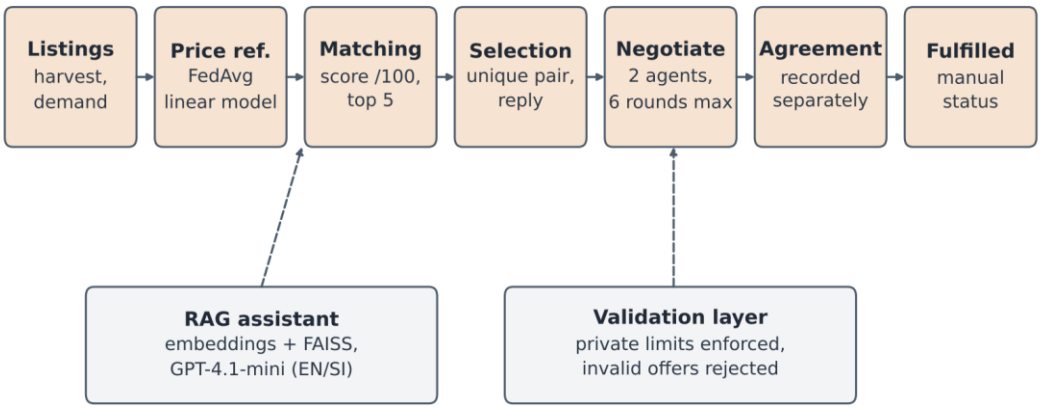


Figure 5 : C03 marketplace transaction life cycle and supporting services

3.6 Component 4 summary : Warehouse Coordination (C04)

Owner. Senarathne S. M. B. V. B. (IT22372444).

Subproblem. SP4, unverifiable stock reports, unprovable capacity claims, undetected anomalous behaviour and unauditable redistribution decisions in the PMB warehouse network.

Input. Stock events and supporting documents from warehouse supervisors; disaster declarations and redistribution decisions from Regional Managers; warehouse registrations from administrators.

Core method. Stock events, disaster declarations, redistribution orders and proof verification results are anchored as Go chaincode assets on a permissioned two-organisation Hyperledger Fabric channel representing PMB headquarters and the Ministry of Agriculture, with CouchDB world state [12]. A Circom circuit using the Groth16 proving scheme [21] lets a warehouse prove that its available capacity meets a threshold without revealing the value to peer organisations or the ledger. A two-layer Graph Attention Network (GAT) [22] computes reliability scores from eight behavioural features propagated across the redistribution graph, following evidence that graph-based detectors capture collusive patterns [23]. A composite score (0.40 distance, 0.35 capacity, 0.25 reliability) ranks evacuation destinations during a disaster, and SHA-256 digests recomputed at verification time detect direct database tampering. Figure 3.5 shows the endorsement and commit path that every anchored transaction follows across the two organisations, and Figure 3.6 the synthetic graph over which reliability scores are propagated.

Output. A role-separated mobile application for administrators, Regional Managers, warehouse supervisors and auditors, and a chronological audit timeline combining relational and on-chain records.

Contribution to the whole. C04 is the storage and distribution layer of the ecosystem and its trust anchor: it makes the national reserve auditable and provides warehouse capacity and reliability information that can be exposed to the rest of the platform through the shared gateway.

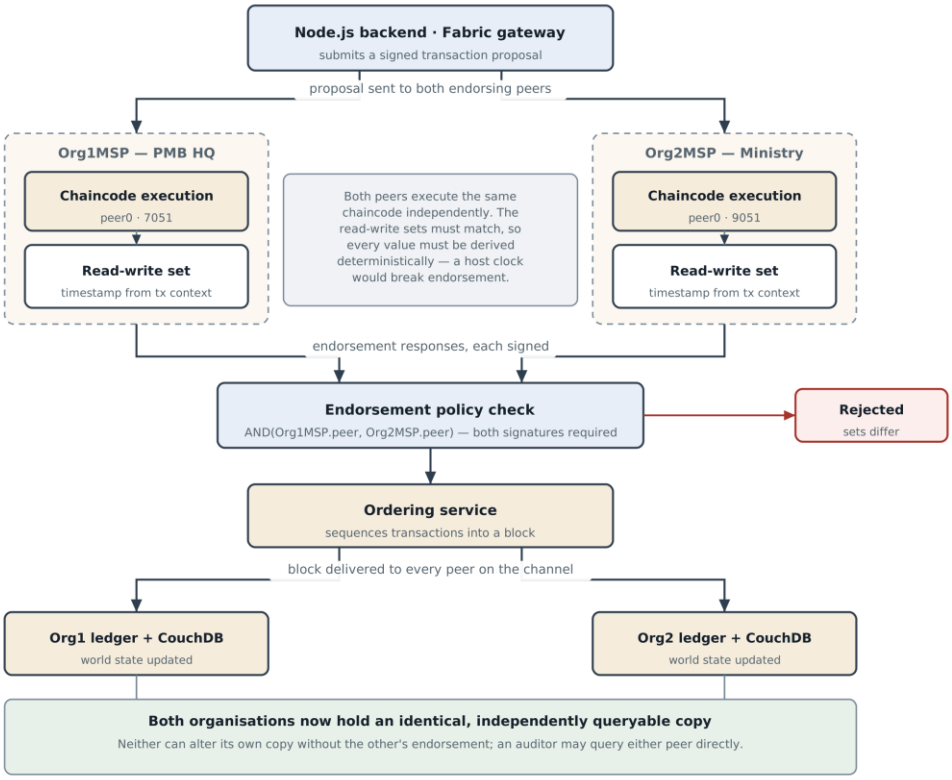


Figure 6 : Hyperledger Fabric transaction endorsement and commit flow in C04

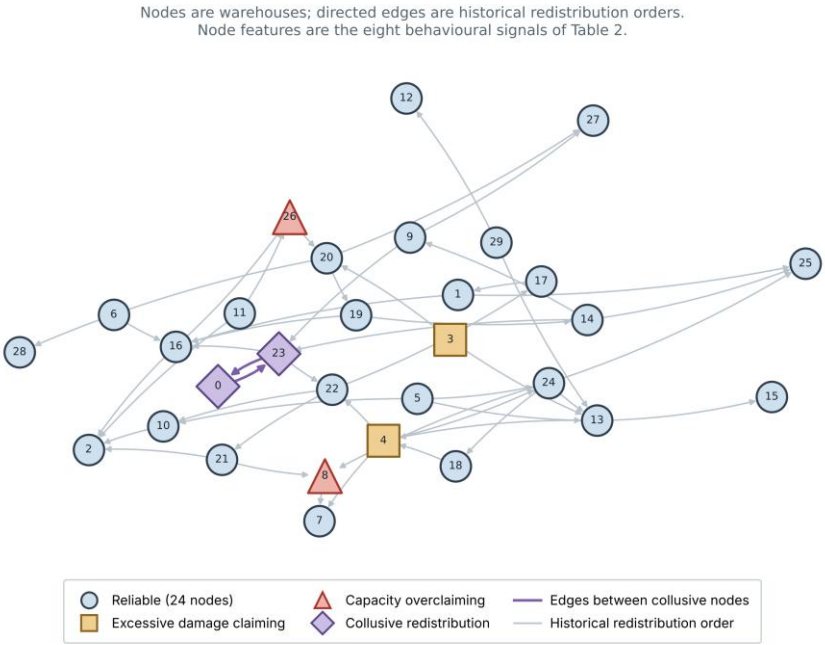


Figure 7 : Synthetic warehouse graph used for C04 reliability scoring, with nodes coloured by injected behaviour

Dimension	C01 Price Forecasting	C02 Digital Farming	C03 Marketplace	C04 Warehouse
Stage	Pricing	Growing	Trading	Storing and distributing
Primary users	Farmers, traders, officers	Farmers	Farmers, millers	PMB staff, auditors
Core AI or crypto technique	XGBoost, SHAP, LLM	ResNet50, XGBoost, Random Forest	Weighted matching, FedAvg, RAG, LLM agents	Fabric chaincode, Groth16 ZKP, GAT
How the output is made trustworthy	SHAP evidence; LLM constrained to evidence	0.85 confidence gate; rule engine	Transparent score; validation of agent offers	Proof verification; ledger anchoring; digest recomputation
Sensitive data protected	Not applicable (public price data)	Farm location and yield (TLS, keyed access)	Negotiation limits, user details	Capacity values, personal data off chain
Backend	Python, FastAPI	Python, FastAPI, Uvicorn	Node.js Express, Python FastAPI	Node.js TypeScript, Python
Data store	CSV and model artefact	Firebase Realtime DB	MongoDB	MySQL and Fabric with CouchDB
Languages in UI	English, simplified Sinhala	English	English, Sinhala	English, Sinhala, Tamil

Table 5 : Side-by-side comparison of the four components

3.7 End to end user and data flow

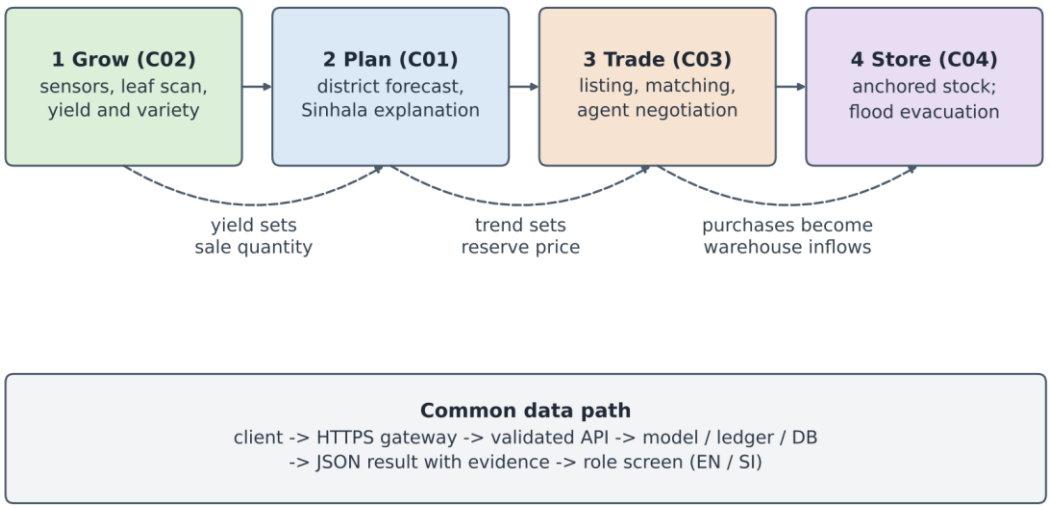


Figure 8 : End-to-end user and data flow across the paddy life cycle

1. **Grow (C02).** During the season the farmer's field node streams temperature, humidity and soil moisture to Firebase. The farmer photographs a suspicious leaf; the diagnosis is shown only if the classifier is at least 85% confident. The yield endpoint combines live telemetry (or weather-API fallback) with location data to estimate kilograms per hectare, and the rule engine issues irrigation or disease alerts.
2. **Plan the sale (C01).** Approaching harvest, the farmer requests a six-week forecast for the district. The response contains weekly prices, the trend class, confidence and risk categories, the most influential features and an explanation in simple Sinhala. The expected yield from C02 and the expected price from C01 together inform how much to sell and at what reserve price.
3. **Trade (C03).** The farmer creates a harvest listing and receives a price reference. Matching ranks compatible miller demands, the farmer selects one, the miller responds, and the two negotiation agents bargain within each party's private limit. An agreement is recorded; the sale is marked as fulfilled only when the user confirms it.
4. **Store and protect (C04).** Paddy purchased by the PMB enters a warehouse as an inflow stock event, hashed and anchored to the ledger. If that warehouse is later flooded, the Regional Manager declares a disaster; candidate warehouses prove sufficient capacity without disclosing it, are ranked by distance, absorbable quantity and reliability, and the chosen redistribution order is executed atomically and anchored for audit.

Across all four stages the technical data path is the same: mobile client, HTTPS boundary, component API with schema validation, model or ledger or database, and a structured JSON response carrying both the result and its supporting evidence.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The group adopted an **interface-first, service-oriented integration strategy**. Early in the project the members agreed the presentation technology (React Native with Expo), the communication style (REST over HTTPS with JSON bodies), the validation approach (schema validation at every API boundary, using Pydantic in the Python services and schema-based parsing in the Node.js services) and the authentication style (JSON Web Tokens). Each member then owned one service end to end, including its data store, its API and its tests, and was responsible for publishing the request and response formats of that API to the others. This sequence reduced the risk that one late component would block the others, because each service could be built and tested in isolation against its documented contract.

Integration proceeded in three stages:

- **Stage 1: component-internal integration.** Each member integrated the internal modules of their own component (for example the C01 data loader, model loader, SHAP and LLM services; or the C04 backend, model service and Fabric gateway) and verified them through component integration tests.
- **Stage 2: client integration.** Each service was connected to its role-based mobile screens and verified through API tools (Postman, Swagger UI) and the client application.
- **Stage 3: ecosystem integration.** Services were exposed behind the shared Digital Goviya HTTPS boundary, and the cross-component data exchanges described in Section 4.2 were defined. The C04 service was hosted in the same cloud region as the integration gateway to minimise the latency between them.

The main integration risks and the way they were managed are summarised in Table 4.1.

Risk	Mitigation adopted
Heterogeneous stacks (Python FastAPI, Node.js Express, TypeScript, Go chaincode)	Integration only through HTTP and JSON; no shared runtime or shared database
Malformed or unexpected payloads breaking ML models	Strict schema validation at each boundary; HTTP 422 or equivalent validation errors returned before model execution
Unavailability of external AI services or field hardware	Fallback paths: weather API for missing telemetry (C02); explanation fallback separate from prediction (C01); graceful handling of unavailable negotiation service (C03)
Mobile devices unable to reach development hosts (localhost)	Backend base URLs supplied through environment configuration (for example `EXPO_PUBLIC_MARKETPLACE_API_URL`); HTTPS public endpoint for C04
Inconsistent identifiers between components (district and paddy-type names)	Agreed shared reference values (Section 4.2); identified exact-string mismatch as a known limitation
Secrets leaking through repositories	Credentials and API keys supplied through environment variables, excluded from source control

Table 6 : Integration risks and mitigations

4.2 Interfaces data contracts and shared models

Each component exposes a small set of resource-oriented endpoints. Table 4.2 summarises the externally relevant interfaces; the full interface specification is given in Appendix A.

Comp.	Interface	Request contract	Response contract
C01	`POST /predict`	district, date	predicted price (Rs./kg)
C01	`POST /forecast`	district, start date, weeks (integer, at least 1)	list of date-aligned weekly prices
C01	`POST /explanation/evidence`	district, date	prediction, previous price, trend, confidence, risk, outlook, recommendations, SHAP contributions
C01	`POST /explanation/llm`	evidence request, language	English and simplified Sinhala explanation, or fallback message

Comp.	Interface	Request contract	Response contract
C02	`POST /predict_disease`	multipart image	disease class and confidence, or "Unknown Type" below 0.85
C02	`POST /predict_yield_production`	District, Season, Zone, Latitude, Longitude, field_id	yield (kg/ha), telemetry source (sensor or fallback), advisories
C02	Variety recommendation	district, zone, season, salinity, iron toxicity	variety code with growth duration and flood tolerance
C03	Marketplace API (Express)	authenticated listing, matching, match-selection, assistant and price-reference requests	listings, ranked matches with score and category, answers, price reference
C03	Negotiation API (FastAPI)	match identifier, party constraints held server side	round history, outcome (agreement or no agreement), agreed price
C04	Warehouses, disasters, documents, audit, ZKP, GNN, users	authenticated JSON per route module	warehouse summaries, ranked candidates, proof results, integrity reports, audit timeline, reliability scores

Table 7 : Principal component interfaces

Shared models and identifiers. Three concepts recur across components and form the shared data model of the ecosystem:

- **District.** Used by C01 as the forecasting key, by C02 as a model feature, by C03 in matching (40 of the 100 match points) and by C04 as a warehouse attribute. The components must therefore use the same district names. C03 matches district strings exactly, so different spellings of the same district would silently reduce a match score; a shared district code list is the agreed remedy.
- **Paddy type or variety.** Used by C03 in matching, by C02 in variety recommendation and by C01 indirectly through its price series. A common variety vocabulary allows a C02 recommendation to flow into a C03 listing.
- **User and role.** Each component enforces roles: farmer and miller in C03; administrator, Regional Manager, warehouse supervisor and auditor in C04; and the authenticated farmer linked to a field identifier in C02. Role is always

derived on the server from the authenticated token, never from client-supplied state.

Versioning, validation and errors. Contracts are validated at the boundary (Pydantic `BaseModel` classes in C01 and C02, schema-based parsing in C04 and request validation in C03). Validation failures return client errors before any model or ledger is touched; authorisation failures return HTTP 403; duplicate submissions such as a second capacity proof for the same warehouse and disaster return a conflict response. The component that owns an endpoint is responsible for its error handling, and the client is responsible for presenting the error in the user's language.

Cross-component data exchanges. Beyond each component's own client interface, the group defined the cross-component exchanges shown in Table 4.3. Their status is reported honestly: the integration evidence available for this report verifies each component's own interfaces and the exposure of C04 data through the gateway, while the automated exchanges between components are specified and supported by the interfaces but have not yet been verified end to end.

Producer to consumer	Data exchanged and purpose	Status
C04 to platform gateway	Warehouse capacity and GAT reliability data exposed for other components	Exposed through the shared API gateway; C04 consumes no data from other components by design
C01 to C03	District weekly forecast and trend as context for listings, price reference and negotiation	Interfaces available on both sides; automated exchange specified, not yet verified end to end
C02 to C03	Expected yield (kg/ha) and recommended variety to pre-fill harvest listings	Specified; manual transfer by the farmer in the current prototype
C03 to C04	Agreed quantities purchased by the PMB as anticipated warehouse inflows	Specified as future work; C04 currently records inflows from supervisors
C02 shared Firebase project	Field telemetry linked to authenticated farmer by field_id	Implemented and verified within C02

Table 8 : Cross-component data exchanges and their status

4.3 Shared nonfunctional requirements

The group agreed a common set of non-functional requirements (NFRs) that every component had to respect, while allowing each to set component-specific targets. Table 4.4 records how each NFR is met.

Requirement	How it is met across the ecosystem
Performance	Models loaded once at service start-up rather than per request (C02 caches ResNet50 and XGBoost in memory; C01 loads the XGBoost artefact; C04 loads the verification key once); responsiveness target of one second for interactive reads
Usability and accessibility	Role-based mobile screens; English and Sinhala in C01 explanations and the C03 interface, and English, Sinhala and Tamil in C04; plain-language evidence rather than raw model output
Security	HTTPS for client traffic (C04 with automatically issued certificates; C02 endpoints over TLS); bcrypt password hashing and JWT authentication (C03, C04) with short-lived access tokens and rotating refresh tokens (C04); rate limiting and security headers (C04)
Privacy	Negotiation limits enforced server side and not exposed to the counterparty (C03); capacity values never written to the ledger (C04); no personal data on chain, consistent with the Personal Data Protection Act No. 9 of 2022 [24]; secrets in environment configuration
Reliability and fault tolerance	Weather-API fallback for failed sensors (C02); LLM explanation fallback independent of prediction (C01); bounded negotiation rounds and termination checks (C03); container restart policies and health checks (C04)
Maintainability	Modular services with single responsibilities; automated unit tests (pytest, Jest); PEP 8 style in Python services; documented contracts
Scalability	Stateless API services that can be replicated; asynchronous ASGI server for C02; C04 verified with 50 concurrent reads without error
Responsible AI	Confidence gating, explanation constrained to evidence, rule-based validation of agent actions, and reliability scores presented as one input among several rather than as verdicts

Table 9 : Shared non-functional requirements and how they are met

4.4 Integrated workflow and exception handling

Under normal operation the integrated workflow follows the four stages of Figure 3.7. The more informative view is how the ecosystem behaves when something fails. Table 4.5 lists the important failure paths, the behaviour implemented and the residual risk.

Failure	Implemented behaviour	Residual risk
ESP32 node loses power or signal (C02)	Backend detects a null Firebase payload and fetches temperature and humidity for the farm coordinates from a weather API, substituting a standard soil-moisture value; prediction continues (reported detection in about 12 ms)	Weather data is regional, not field-level; stale (non-null) readings are not yet detected
Uncertain or irrelevant image (C02)	Diagnosis withheld below 0.85 confidence; "Unknown Type" returned	Threshold not yet evaluated on a separate unknown-image set
Unsupported district (C01)	District validated before feature generation; clear "unsupported district" error	Supported-district list must match other components
LLM provider unavailable (C01, C03)	C01 returns structured evidence with a fallback message; C03 assistant reports failure without inventing an answer	Explanation quality depends on provider availability
Agents fail to agree or propose invalid offers (C03)	Validation rejects offers outside private limits; negotiation ends after six rounds with no agreement	Agent behaviour varies with model responses
Duplicate match selection (C03)	Unique database constraint on harvest-demand pair	None significant
Capacity below threshold (C04)	Witness generation fails; no proof can be produced or submitted	Proof attests the reported value, not physical stock
Ledger peer unavailable (C04)	Event committed to the database; anchoring failure logged; unanchored events visible through the audit endpoint	Silent divergence until audited; transactional outbox identified as the fix
Concurrent redistribution orders (C04)	Both warehouse rows locked in sorted identifier order within one transaction	None significant

Table 10 : Failure paths, recovery behaviour and residual risk

The common pattern is that failures in auxiliary services (sensors, language models, ledger peers) never corrupt the primary result: the prediction, the match or the stock record is preserved, and the degraded part is reported explicitly. The most important open weakness is the C04 fire-and-forget anchoring, which can leave a stock event without a ledger record while the client receives a success response; this is discussed further in Section 6.6.

4.5 Team contribution and dependency matrix

Each member was individually responsible for one component, and all four contributed to the shared conventions, the integration and this common report. Table 4.6 maps students to their research, development, integration and evidence responsibilities, and Table 4.7 shows the dependencies between components. A signed contribution declaration is provided in Appendix B.

Student	Research and development	Integration work	Test evidence and shared outputs
Ransara N.S (IT22102096), C01	Price dataset preparation; XGBoost model integration; recursive weekly forecasting; SHAP and XAI modules for trend, confidence, risk, outlook and recommendations; evidence-constrained LLM explanation in English and simplified Sinhala	FastAPI REST service for the Digital Goviya client; district validation fix; LLM fallback	47 XAI unit tests and 5 integration tests passed; latency and repeated-request tests; forecasting sections of this report
Kumarasingha P.A.N.D. (IT22294784), C02	ResNet50 transfer learning with confidence gate; XGBoost yield regression; Random Forest variety recommender; rule engine; ESP32 firmware and sensor calibration	Asynchronous FastAPI backend with in-memory model caching; Firebase telemetry linked to authenticated users; weather-API fallback; CORS and Pydantic contracts	Model comparisons; fallback test; schema validation tests; farmer validation instrument; farming sections of this report
Chamudi K.S.I. (IT22147950), C03	Weighted matching algorithm; FedAvg-style price reference; multilingual RAG assistant; two-agent negotiation with validation	Expo client, Express API and MongoDB; FastAPI negotiation service; match-selection and transaction life cycle; bilingual interface	Price-reference evaluation; matching and negotiation test plan; marketplace sections of this report
Senarathne S. M. B. V. B. (IT22372444), C04	Fabric network and Go chaincode; Groth16 capacity circuit; GAT reliability model; composite redistribution ranking; integrity verification	Node.js and TypeScript backend; role-separated trilingual Expo client; containerised cloud deployment over HTTPS; exposure of capacity and reliability through the gateway	80 unit tests and 12 system tests passed; NFR measurements; data consistency audit; warehouse sections of this report

Table 11 : Summary of each student's contribution

Depends on	C01	C02	C03	C04
C01	-	None	None	None
C02	None	-	None	None
C03	Optional: forecast as price context	Optional: yield and variety to pre-fill listings	-	None
C04	None	None	Future: PMB purchase volumes as inflows	-

Table 12 : Component dependency matrix (row depends on column)

The matrix shows that the ecosystem has no mandatory circular dependencies: every component can operate alone, and integration adds value without creating a single point of failure. C03 is the natural point of convergence, because it is where farm output and price expectations become trade.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The group standardised on a small number of technologies at the presentation and integration layers while allowing each component to choose the analytical stack best suited to its research problem. Table 5.1 lists the shared and component-specific technologies and the reasons for the major common choices.

Layer	Technology	Reason for choice and where used
Mobile client	React Native with Expo and Expo Router	One codebase for Android and iOS; file-based role routing; used by all components
API style	REST over HTTPS, JSON	Language-neutral contracts between Python, Node.js and TypeScript services
Python services	FastAPI on Uvicorn (ASGI), Pydantic	Asynchronous request handling for ML inference; automatic validation and documentation; C01, C02, C03 negotiation, C04 model service
Node.js services	Express (JavaScript in C03, TypeScript in C04)	Mature HTTP ecosystem; ORM and Fabric gateway SDK support
Machine learning	XGBoost, scikit-learn, TensorFlow and Keras, PyTorch Geometric, SHAP	XGBoost in C01 and C02; ResNet50 in C02; FedAvg-style SGDRegressor in C03; GAT in C04
Language and retrieval	OpenAI GPT-4.1-mini, local Ollama model, sentence-transformer embeddings, FAISS	Grounded assistance and negotiation in C03; evidence-to-text explanation in C01
Data stores	Firebase Realtime Database, CSV and model artefacts, MongoDB, MySQL	Store chosen to fit each component's data shape
Trust layer	Hyperledger Fabric with CouchDB, Go chaincode, Circom and snarkjs (Groth16)	Permissioned multi-organisation audit trail and zero-knowledge capacity proofs in C04
Edge hardware	ESP32, DHT22, capacitive soil-moisture probe, Arduino C++ firmware	Low-cost field telemetry in C02
Packaging and hosting	Docker and Docker Compose; Caddy reverse proxy; Oracle Cloud Infrastructure ARM instance	Reproducible multi-service deployment with automatic TLS for C04

Layer	Technology	Reason for choice and where used
Testing tools	pytest, Jest, Postman, Swagger UI	Automated unit tests and API-level integration and system tests

Table 13 : Common and component-specific technology stack

5.2 Development and integration process

Development followed an iterative, interface-first process. After the project proposal and the agreement of shared conventions, each member worked in short iterations on their own service, demonstrating progress to the supervisors at regular review meetings and adjusting scope in response to feedback. Interfaces were documented before the corresponding screens were built, so that client and service work could proceed in parallel.

Source code for each component was kept under version control, with secrets excluded and supplied through environment configuration. Each service carries its own automated tests (pytest for the Python services and Jest for the C04 backend), so that a change can be checked by re-running the suite from the command line. Defects found during integration were fixed and re-tested: for example, the C01 district-validation order was corrected so that unsupported districts are reported clearly, and the C04 chaincode timestamp source was changed to the transaction context after non-deterministic timestamps broke multi-organisation endorsement.

A notable practice adopted by the C04 member, and recommended for the whole ecosystem, was a **data consistency audit** before release: querying the stored data directly to reconcile derived values against event history. This surfaced three high-severity defects that the passing test suites had not detected (Section 6.2), and shows that integration testing should verify the meaning of stored data and not only the shape of API responses.

5.3 Deployment and operational configuration

The ecosystem is deployed as independent services, each configured through environment variables for its database connection, model artefacts, external API credentials and public base URL.

- **C04 warehouse coordination** is deployed to an Oracle Cloud Infrastructure VM.Standard.A1.Flex instance (two ARM cores, 12 GB memory, Ubuntu 24.04). Nine containers run on the instance: a Caddy reverse proxy that terminates TLS with automatically renewed certificates, the backend, the GAT model service, MySQL, a Fabric ordering service, two peers and two CouchDB instances; each peer also starts a chaincode container at runtime. Only ports 80 and 443 are exposed. The backend binds to the loopback interface and joins both the application and Fabric container networks so that it reaches the peer by the DNS name on its certificate. Containers restart automatically on failure.
- **C02 digital farming** runs as a FastAPI application served by Uvicorn (for example `uvicorn app:app --host 0.0.0.0 --port 8000`) on a cloud host reachable by the mobile client over HTTPS. All model binaries and the variety catalogue are loaded into memory at start-up, so server memory is monitored. Field nodes are sealed in weather-resistant housings and powered by batteries with small solar panels; Firebase read and write quotas are monitored.
- **C01 price forecasting** runs as a FastAPI service that loads the cleaned historical dataset and the serialised XGBoost model, with the LLM provider credential supplied through the environment. Model and library versions must match, because a serialised model loaded under a different XGBoost version may raise compatibility warnings.
- **C03 marketplace** runs as an Express backend (port 5000) with MongoDB, a FastAPI negotiation service (port 8000) and Python RAG and price-reference modules. The RAG module requires its knowledge base, FAISS index and OpenAI credential; the negotiation service requires a running Ollama model. The mobile client reaches the backend through `EXPO_PUBLIC_MARKETPLACE_API_URL`.

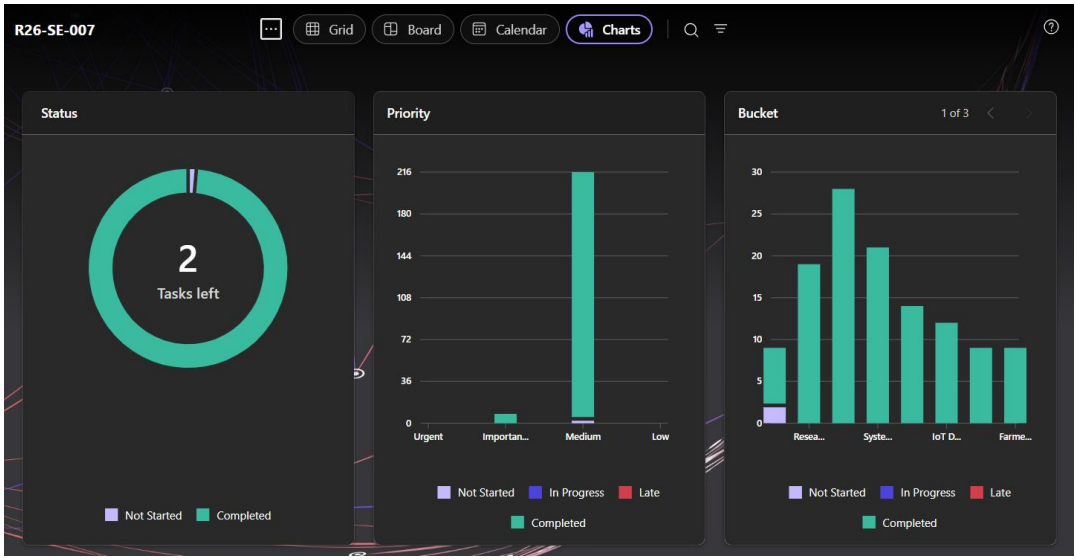
Monitoring, backup and restore. Health checks and container restart policies provide basic recovery for C04. Operational logs record failed requests and unanchored ledger events while avoiding passwords, API keys and private negotiation limits. The authoritative recovery path for warehouse records is the Fabric ledger, which is replicated on both peers; relational data and model artefacts should be backed

up regularly, and models must be re-evaluated before a retrained version is deployed. A production deployment would add centralised monitoring and scheduled database backups, which were outside the prototype's scope.

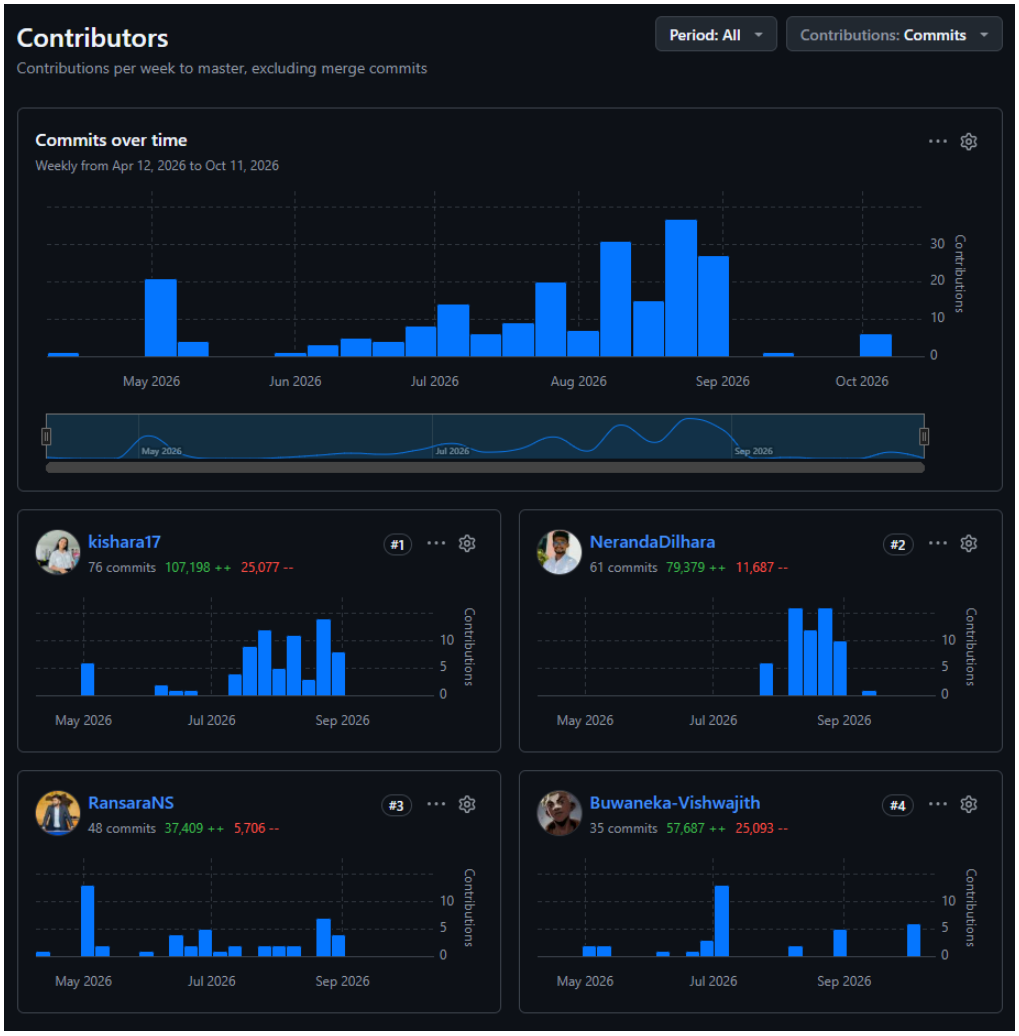
5.4 Evidence of team execution

Execution was coordinated through supervisor review meetings, shared team communication channels and each member's version-controlled repository and research records. The principal evidence of execution is summarised below; raw activity logs are retained by the group and are available on request rather than reproduced here.

- **Planning and tracking.** Tasks for the proposal, progress presentations, final implementation and reports were planned and tracked by the group.



- **Version control.** Each component has its own repository with commit history covering implementation, tests and documentation.



- **Meetings and research diaries.** Regular meetings with the supervisor, co-supervisor and external supervisor reviewed progress and direction; each member kept a research diary of decisions and experiments.
- **Delivered artefacts.** Four working services with automated tests, role-based mobile clients, a public HTTPS deployment of the warehouse stack, four individual theses and this common report.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

Because the four components perform different tasks, no single accuracy figure can describe the ecosystem. The group therefore defined group-level evaluation questions (EQ) that follow from the group objectives, and allowed each component to answer them with metrics appropriate to its task. The framework distinguishes four kinds of evidence, which must not be confused: **functional correctness** (does the software do what it is specified to do), **model performance** (how accurate are the predictions on the data available), **operational quality** (latency, reliability, security) and **stakeholder value** (do users find it useful). Table 14 summarises the framework.

EQ	Evaluation question	Metrics and acceptance criteria	Data and method
EQ1	Does each stage provide useful AI decision support? (GO1)	Classification accuracy and loss compared with baselines (C02); MAE, RMSE, R^2 and MAPE (C01, C02, C03); rule correctness (C03); zero false acceptances and in-sample fit against a majority baseline (C04)	Held-out validation sets; synthetic data where real data was unavailable; controlled scenarios
EQ2	Is every output explained or verifiable? (GO2)	Presence and correctness of evidence fields; confidence gate behaviour; integrity checks detect tampering	Unit, integration and system tests
EQ3	Is sensitive data protected? (GO3)	No negotiation limit or capacity value disclosed to counterparties or the ledger; role access enforced	Ledger inspection; role-based access tests; design review against threat model
EQ4	Is access usable and bilingual? (GO4)	Interfaces implemented for each role; English and Sinhala outputs	Functional review; planned user study
EQ5	Can components be composed through their interfaces? (GO5)	Endpoints validate contracts and return documented responses; deployment reachable over HTTPS	Postman and Swagger tests; deployment verification
EQ6	Is operation responsive and fault tolerant? (GO6)	Interactive responses under 1 s; 100% success in repeated-request tests; continued prediction under sensor failure	Timed requests; repeated and concurrent requests; hardware disconnection test

Table 14 : Group-level evaluation framework

6.2 Integration and end to end testing

Testing was carried out at the unit, integration, system, regression, performance and security levels. Table 6.2 consolidates the executed tests and their recorded outcomes across the ecosystem. Only tests whose outcomes are recorded in the individual theses are reported as executed; planned tests are identified as such.

Comp.	Level	Scope	Recorded outcome
C01	Unit	Trend, confidence, feature importance, recommendation and summary modules	6 trend tests passed; combined XAI run of 47 tests passed
C01	Integration	Model loading, multi-week forecast, structured evidence endpoint, LLM explanation endpoint	1, 2, 1 and 1 tests passed respectively
C01	System and regression	Valid requests, unsupported district, insufficient history, zero-week horizon	District validation defect found, fixed and re-tested; zero-week request rejected
C01	Reliability	20 consecutive and 30 sequential prediction requests	20/20 and 30/30 successful
C02	Model validation	ResNet50 against MobileNetV2 and EfficientNetB0; XGBoost against Decision Tree and Random Forest	ResNet50 and XGBoost selected (Section 6.3)
C02	API	Mock JSON and image payloads; missing parameters; concurrent requests	Correct HTTP status codes, including 422 for missing fields; no event-loop blocking observed
C02	Fault tolerance	ESP32 power disconnected during operation	Fallback to weather API triggered; yield prediction completed
C03	Model evaluation	Price-reference model on 1,000 samples	MAE 11.55, RMSE 12.27, R^2 0.18, MAPE 9.53%
C03	Functional and integration	Authentication, listings, matching, duplicate selection, RAG, price, negotiation, authorisation, status update, service failure (TC01 to TC12)	Test plan defined; execution evidence to be recorded per case
C04	Unit	Haversine distance, ranking, JWT, document hashing, feature extraction, model inference	80 of 80 tests passed (50 Jest, 30 pytest)

Comp.	Level	Scope	Recorded outcome
C04	System (deployed)	ST-01 to ST-12: capacity validation, anchoring, integrity under tampering, proof below threshold, duplicate proof, ranking, atomic redistribution, stock conservation, auditor and role access	12 of 12 passed
C04	Security and access	Unauthenticated requests, cross-facility access, administrator endpoints as supervisor	Rejected as specified
C04	End to end	Stock event through public HTTPS endpoint to ledger and back	Digests matched across database, recomputation and ledger on both peers

Table 15 : Consolidated test evidence across components

End-to-end testing. Two levels of end-to-end testing were achieved. Within components, complete workflows were exercised from the client or API tool through the service to its store and back: for C04 against the publicly deployed instance, for C02 from image or sensor input to advisory payload, and for C01 from request to bilingual explanation. Across components, the shared conventions of Chapter 4 were verified by the fact that every service accepted and returned the agreed JSON formats over HTTP. A full scripted scenario that drives one farmer's data automatically through C02, C01, C03 and C04 was specified (Figure 3.7) but has not yet been executed as a single automated test, and this is recorded as a limitation in Section 6.6.

What testing revealed. The most valuable integration findings came from failures. The C04 data consistency audit found that integrity digests had been computed with an unpersisted timestamp and could never be reproduced; that early redistribution orders moved stock towards a flooded warehouse rather than out of it, so seven orders totalling 152 tons had moved no stock at all; and that a production build had never succeeded because the development runner skipped type checking. None of these were caught by unit or integration tests, because each concerned the meaning of stored data rather than the behaviour of a single function. The C01 tests revealed that an unsupported district was misreported as insufficient history. These findings informed the group recommendation in Section 8.3 that reconciliation checks become part of the integrated test suite.

6.3 Overall results

Table 16 synthesises the principal quantitative results of the four components. Detailed experiments are presented in the individual theses.

Comp.	Capability	Principal result	Interpretation and evidence status
C01	XGBoost price prediction	RMSE 1.25, MAE 0.84, MAPE 0.71% in the notebook evaluation	Promising; split and date alignment need documenting before treating as validated forecast accuracy
C01	Six-week Anuradhapura forecast from 31 Aug 2026	Weekly values from 112.07 to 111.91 (Rs./kg)	Demonstrates recursive forecasting; accuracy against actual prices not yet measured
C01	API latency (10 requests each)	/predict 53.3 ms, /forecast 105.7 ms, /explanation/llm 371.0 ms average; 100% success	Measured in test environment
C02	Disease classification	ResNet50 validation accuracy 90.91% (loss 0.2075), against 86.45% for MobileNetV2 and 88.10% for EfficientNetB0	Reported results; raw training logs to be attached
C02	Yield regression	XGBoost R^2 0.9890, MAE 34.20 kg/ha, RMSE 48.15 kg/ha, against Random Forest R^2 0.9425 and Decision Tree R^2 0.8410	Training data augmented with synthetic environmental features
C02	Sensor failover	Fallback detected and triggered in about 12 ms; prediction completed	Single reported observation; repeated trials needed
C03	Price reference (FedAvg simulation)	MAE 11.55, RMSE 12.27, R^2 0.18, MAPE 9.53%; mean predicted Rs. 122.86/kg against actual Rs. 124.81/kg	Low R^2; synthetic data and simulated clients; baseline only
C03	Matching and negotiation	Score out of 100 with three categories; negotiation bounded to six validated rounds	Rules implemented; trading success not yet measured
C04	Zero-knowledge capacity proof	Zero false acceptances on all invalid inputs; proof generation 2 to 5 s; 38 constraints	Constraint enforcement demonstrated

Comp.	Capability	Principal result	Interpretation and evidence status
C04	GAT reliability scoring	26 of 30 nodes correct in sample (BCE loss 0.2989 at epoch 200)	Only two above the 24-node majority baseline; not a detection claim
C04	Integrity and audit	Direct database tampering detected by digest recomputation; tampering reversed restores integrity	Demonstrated on deployed system
C04	Latency and scalability	Reads 310 to 480 ms; writes with anchoring 640 to 910 ms; 50 concurrent reads without error	Includes about 250 ms round trip from Sri Lanka to the US East region

Table 16 : Synthesised quantitative results

Figure 6.1 compares the C02 models against their baselines, showing the gains that justified selecting ResNet50 and XGBoost. Figure 10 compares the measured response times of the C01 and C04 endpoints against the one-second responsiveness target agreed for interactive use.

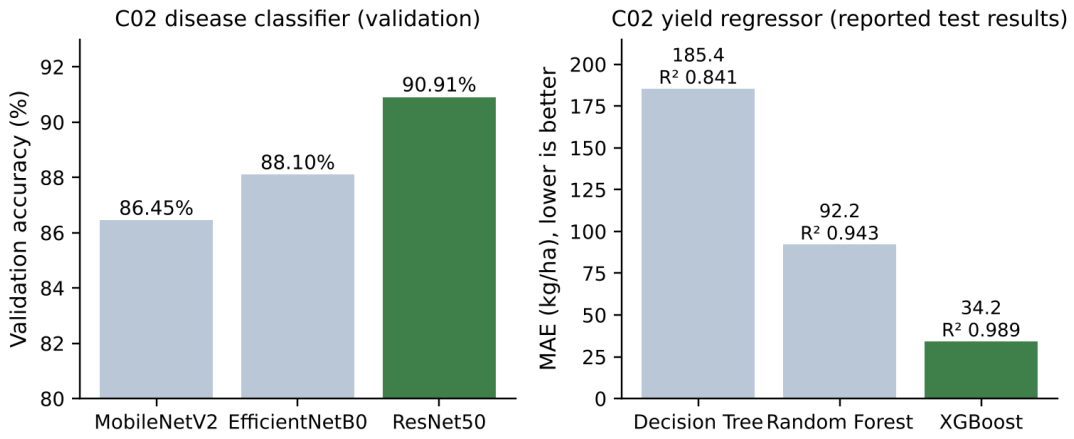


Figure 9 : C02 model selection results: disease classifier accuracy and yield regressor error

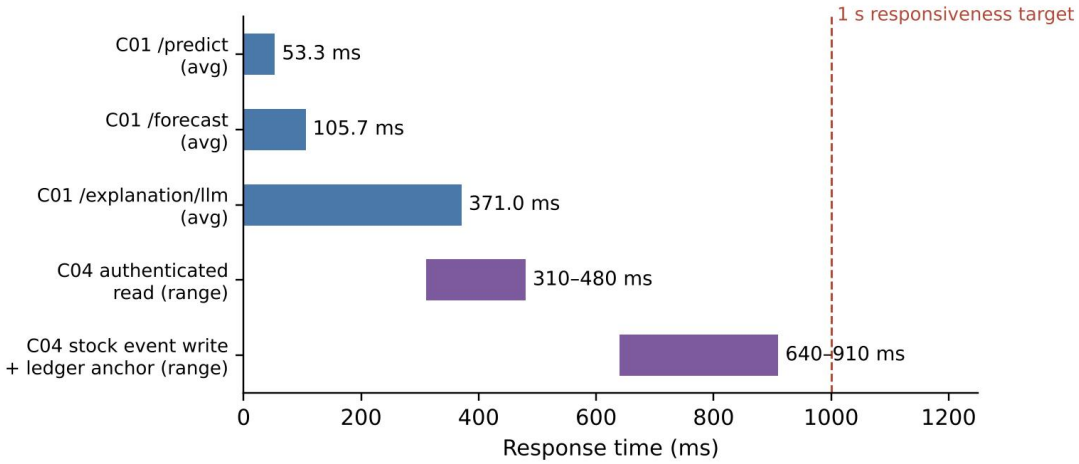


Figure 10 : Measured API response times for C01 and C04 against the one-second target

Taken together, the results show a consistent pattern. Every component delivers a working AI or cryptographic mechanism with evidence attached to its output, and every measured interactive endpoint responds within the one-second target, including the C04 write path that commits to a two-organisation blockchain. The strength of the predictive evidence varies: the C02 models and the C01 notebook evaluation report strong fit on their datasets; the C03 price reference and the C04 anomaly detector are honest baselines on synthetic data. The components whose guarantees are structural rather than statistical (the C02 confidence gate, the C03 validation layer, the C04 proof circuit and integrity check) are the most firmly established.

6.4 Stakeholder and user validation

Who evaluated the solution. The solution was reviewed throughout development by the project supervisor, Dr. Mahima Weerasinghe, the co-supervisor, Mr. Eishan Weerasinghe, and the external supervisor, Mr. Suranga Senanayake, whose knowledge of PMB warehouse operations supplied the C04 requirements, including the observation that coordination currently relies on telephone calls and messaging. The C04 system was demonstrated with a Regional Manager role and realistic warehouse and disaster scenarios.

How evidence was collected and what it showed. Supervisor and expert reviews shaped several design decisions. The most significant was the correction of the C04 redistribution direction, from moving stock into an affected warehouse to evacuating

it, which aligned the ranking metric with operational practice. Reviews also reinforced the group's decision to present all AI outputs as decision support and to report limitations explicitly.

Formal user studies. A structured farmer validation study was designed for C02: ten paddy farmers, hands-on use of disease detection, yield prediction, variety recommendation and environmental monitoring, and a five-point Likert questionnaire covering ease of use, interface clarity, usefulness of each feature and overall satisfaction, with voluntary participation, informed consent and anonymised reporting. Similar task-based evaluations with farmers and millers were planned for C03 and with farmers and agricultural officers for C01. At the time of writing, completed questionnaire results from farmers, millers or PMB staff are not available, and the illustrative values presented in the C02 thesis are a reporting template, not collected data. **User acceptance of the integrated solution has therefore not yet been empirically established**, and this report makes no claim about it.

Remaining concerns. Stakeholder-facing concerns raised during review include the risk that a low C04 reliability score could be read as an accusation against a supervisor, which the interface addresses by presenting the score as one ranking input among three and by recording no automated action; and the risk that farmers might treat a price forecast or price reference as a guaranteed price, which the interfaces address by labelling outputs as estimates.

6.5 Achievement of common objectives

Table 17 maps the evidence to each group objective and states whether it was fully, partly or not achieved.

ID	Status	Evidence	Justification
GO1	Achieved (technical); partly (real-world)	All four stages have working, evaluated AI support (Table 6.3)	Strong reported metrics for C02 and C01; C03 and C04 learning models are baselines on synthetic data
GO2	Achieved	SHAP and XAI evidence with constrained LLM text (C01); confidence gate and rule engine (C02); transparent score and validated negotiation (C03); proofs, anchoring and digest recomputation (C04)	Every component returns evidence with its output, verified by tests
GO3	Achieved within the stated threat models	Limits enforced server side (C03); capacity never on ledger, no personal data on chain (C04); TLS and role-based access	Guarantees are scoped: the C04 proof establishes that a capacity claim clears a threshold, not that the claim is accurate; C04 confidentiality excludes the application server; C03 separation is application-level, not cryptographic
GO4	Partly achieved	Role-based mobile clients for all components; Sinhala in C01, C03 and C04, Tamil in C04	Sinhala output not yet uniform across all components; usability not yet measured with users
GO5	Partly achieved	All services expose validated REST/JSON interfaces; C04 deployed over HTTPS and exposed through the gateway	Automated cross-component exchanges specified but not yet verified end to end
GO6	Achieved under test conditions	All measured interactive endpoints under 1 s; 100% success in repeated-request tests; sensor fallback; container recovery	Not yet load-tested at national scale

Table 17 : Achievement of group objectives

6.6 Integrated limitations and threats

Data limitations. The most pervasive limitation is the absence of real operational data at three of the four stages. C02 augmented historical yield records with synthetic temperature and soil-moisture features; the C03 price reference was trained on 1,000 synthetic records across simulated clients; and the C04 GAT was trained and evaluated in sample on a 30-node synthetic graph whose anomaly ranges were chosen to be separable. These results establish that the mechanisms work, not that they generalise.

Integration limitations. The components integrate through common conventions and documented interfaces, but the automated data exchanges between them (Table 4.3)

are not yet verified end to end, and authentication is not yet unified: C02 uses the shared Firebase authentication project, while C03 and C04 issue their own JWTs against their own user stores. A user therefore needs separate accounts per component in the current prototype. Shared identifiers such as district names are agreed but not enforced by a common code list.

Dependency risks. C01 explanations and C03 assistance depend on external LLM providers; C02 depends on Firebase quotas and a weather API; C04 depends on a single-node ordering service. The C04 anchoring path can silently diverge from the database when the ledger is unavailable.

Threats to validity. Construct validity is limited where synthetic generators encode assumptions about real behaviour (C04 anomaly patterns, C02 environmental distributions). Internal validity is limited where evaluation is in sample (C04) or where raw logs and splits are not yet attached (C02, C01). External validity is limited by the small scale of tests (five operational warehouses in C04; one forecast scenario in C01) and by the absence of field trials across seasons and districts. Conclusion validity is limited by the lack of user-study data.

Generalisability. The architectural pattern (independent AI services, evidence-carrying outputs, privacy-scoped trust mechanisms and API-level integration) is not specific to paddy, and the C04 coordination design applies to any multi-agency storage network. The specific models, however, must be retrained and re-evaluated on real Sri Lankan data before any operational use.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

Digital Goviya offers different value to each participant, but the common thread is **better decisions on trustworthy information without unnecessary disclosure**.

Beneficiary	Value created	Supporting evidence
Paddy farmers	Confident disease diagnosis and early alerts; yield and variety advice tied to current field conditions; explained price outlook in Sinhala; easier access to millers with structured negotiation	C02 accuracy and fallback results; C01 XAI and bilingual output; C03 matching and negotiation workflow
Rice millers	Ranked suppliers that match district, type, price and quantity; negotiation that respects private limits	C03 weighted score and validation layer
Paddy Marketing Board	Verifiable stock records; capacity checks without disclosure; ranked evacuation options; audit trail independent of the application database	C04 system tests, integrity results and deployment
Ministry of Agriculture and auditors	Read-only, tamper-evident view of national reserve decisions	C04 two-organisation channel and auditor role
Agricultural officers and policy makers (future)	Aggregated disease, yield and price signals by district	C02 and C01 data, subject to consent and aggregation
Society	Contribution to food security, reduced post-harvest loss and more transparent commodity markets	SDG alignment (Section 7.5)

Table 18 : Value proposition by beneficiary

7.2 Market alternatives and differentiation

Target segments. The ecosystem addresses three segments: (1) smallholder paddy farmers and farmer organisations, reached through a free or low-cost mobile application; (2) rice millers and paddy collectors seeking reliable supply; and (3) public-sector and cooperative storage operators, beginning with the PMB, for whom C04 is procured as operational software.

Existing alternatives. Farmers currently rely on extension officers, informal networks and general-purpose mobile advisory services. Crop diagnosis applications such as

Plantix provide image-based diagnosis but not paddy-specific yield, variety, price or market functions [10]. Published price bulletins report past prices without forecasts or explanations. Farmer-miller trade is arranged by telephone and through intermediaries. Warehouse coordination uses telephone calls, messaging and conventional databases, and agricultural blockchain systems focus on traceability rather than inter-facility coordination [25], [9]. Privacy-preserving ledger auditing has been demonstrated in finance [26] but not for warehouse capacity claims.

Differentiation. Table 19 compares Digital Goviya with these alternatives on the capabilities that matter to the common problem.

Capability	Diagnosis apps	Price bulletins	Informal trading	Warehouse databases	Digital Goviya
Paddy disease diagnosis with confidence gate	Partial	No	No	No	Yes
IoT-driven yield and variety advice with fallback	No	No	No	No	Yes
Explained multi-week price forecast in Sinhala	No	No	No	No	Yes
Ranked farmer-miller matching and constrained negotiation	No	No	No	No	Yes
Tamper-evident, multi-organisation stock audit trail	No	No	No	No	Yes
Capacity verification without disclosure	No	No	No	No	Yes
One mobile ecosystem across the supply chain	No	No	No	No	Yes

Table 19 : Differentiation against existing alternatives

The defensible differentiation lies less in any single algorithm, since each technique is established in the literature, than in the **combination**: explained and verifiable AI at every stage, privacy scoped to each stakeholder's real concern, and a common integration layer, applied specifically to the Sri Lankan paddy chain.

7.3 Business model and feasibility

Revenue and sustainability model. A hybrid model is proposed, recognising that farmers have limited ability to pay while institutions have both the need and the budget:

- **Farmers (free core service).** Disease diagnosis, price outlook and basic marketplace access are free, to maximise adoption and the social value of the platform. IoT field kits can be sold or leased through farmer organisations or cooperatives, with optional premium advisory services.
- **Millers and collectors (subscription or success fee).** Access to ranked supplier matching and negotiation support is offered through a monthly subscription or a small fee on agreements that reach fulfilment.
- **PMB and storage operators (institutional licence).** C04 is offered through a development and maintenance contract, as is typical for government operational software, followed by a per-facility subscription: an indicative Standard tier of LKR 4,500 per facility per month (operational recording, role separation and document integrity) and a Verified tier of LKR 12,000 per facility per month (adding ledger anchoring, capacity proofs, reliability scoring and audit export), priced to reflect the cost of running blockchain peers.
- **Public-sector partnerships.** Aggregated, consented and anonymised district indicators of disease incidence, expected yield and price trends could support the Department of Agriculture and agricultural insurers, subject to the data protection constraints in Section 7.4.

Channels and partners. The PMB and the Ministry of Agriculture (institutional customers and data partners), farmer organisations and agrarian service centres (onboarding and kit distribution), telecommunication providers (mobile reach and data bundles) and agricultural research institutes (validation and data).

Key resources and costs. Cloud hosting, LLM and weather API usage, field hardware, Firebase usage, model retraining, support and maintenance. The C04 deployment demonstrated that the entire warehouse stack, including a two-organisation Fabric

network, runs on a two-core ARM free-tier cloud instance, which materially lowers infrastructure cost. The ESP32 field node uses low-cost commodity sensors.

Technical and operational feasibility. All four components were implemented and tested, and the most infrastructure-intensive one was deployed publicly over HTTPS and verified end to end, which demonstrates technical feasibility. Operational feasibility depends on the PMB's willingness to adopt a new coordination process, on farmer onboarding and connectivity, and on access to real data for retraining, which are the subject of the adoption plan below.

7.4 Adoption intellectual property and compliance

Adoption barriers and responses. Digital literacy and trust are the main barriers for farmers, addressed through Sinhala interfaces, plain-language explanations, confidence gating and onboarding through farmer organisations. For millers, the barrier is reluctance to change established relationships, addressed by making matching advisory rather than exclusive. For the PMB, procurement rules and change management are the barriers; the containerised architecture supports hosting on government infrastructure if required, and the standard tier allows adoption before the cryptographic layer.

Ownership and licensing. The software was developed as academic work at the Sri Lanka Institute of Information Technology. The group grants the Institute a non-exclusive right to reproduce and distribute the report, and ownership and licensing of the software for any commercial use would be agreed with the Institute and any partner organisation before deployment. Open-source dependencies are used under their respective licences, and model providers' terms of use apply to the LLM services.

Data protection. The Personal Data Protection Act No. 9 of 2022 imposes obligations of lawful basis, purpose limitation and data subject rights on controllers of personal data [24]. The design responds in three ways: personal data is never written to the immutable ledger, so the right to erasure is not obstructed; farm coordinates and telemetry are transmitted over TLS and linked to authenticated users; and negotiation limits and capacity values are not disclosed to other parties. Any aggregation of farmer data for public-sector use would require explicit consent and anonymisation.

Ethics, safety and professional responsibility. AI outputs could cause harm if over-trusted: a wrong diagnosis may lead to unnecessary chemical use, a misleading forecast may influence a sale, and a low reliability score may damage a supervisor's reputation. The ecosystem therefore withholds uncertain diagnoses, constrains LLM explanations to model evidence, labels prices as estimates, separates agreements from completed sales and presents reliability as one input with no automated sanction; a defined appeal process would be required before reliability scores influence staffing decisions. Use of generative AI in preparing the theses and this report is disclosed in the declaration.

7.5 Sustainable Development Goal contribution

Against the targets identified in Section 1.3, the plausible contributions of the ecosystem are:

- **SDG 2.3 and 2.4.** Earlier and more confident disease diagnosis and condition-aware irrigation advice can reduce yield loss and avoid unnecessary chemical application; better market access and price awareness can improve smallholders' bargaining position. These effects are plausible but unmeasured.
- **SDG 2.c.** Explained price forecasts, transparent matching and a verifiable record of national reserves directly support well-functioning commodity markets and timely information on food reserves; this is the most direct contribution of the integrated solution.
- **SDG 12.3.** Faster evacuation of stock from flooded warehouses, ranked by distance, capacity and reliability, and early disease response in the field target post-harvest and production losses.

Possible negative effects. A digital platform can widen the gap between connected and unconnected farmers; automated negotiation could disadvantage less experienced users if the agents are poorly calibrated; reliance on cloud and AI services carries environmental and financial cost; and data concentration creates privacy and power risks.

Measures needed to sustain benefits. Assisted access through agrarian service centres and farmer organisations; transparent, user-adjustable negotiation limits;

periodic retraining and re-evaluation on real data; independent audit of the reliability model before operational use; and measurement of impact through baseline and follow-up indicators such as post-harvest loss, price received relative to district averages and time to sale.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

This project set out to address a common problem: that decisions along Sri Lanka's paddy supply chain are taken on information that is fragmented, unverifiable and unexplained, while much useful information is too sensitive to share. The evidence presented in this report supports the conclusion that an integrated digital ecosystem can address this problem at the architectural and technical level.

Digital Goviya delivers working AI support at all four stages of the chain. At the farm, a ResNet50 classifier reached 90.91% validation accuracy while refusing to diagnose when less than 85% confident, and an XGBoost regressor predicted yield with an R^2 of 0.989 and continued operating when field sensors were disconnected. At the market-information stage, an XGBoost forecasting service produced district weekly forecasts with SHAP-based evidence and bilingual explanations in well under a second. At the trading stage, the marketplace combined transparent matching, a price reference, grounded knowledge assistance and negotiation bounded by private limits. At the storage stage, a deployed blockchain-backed system enforced capacity proofs with zero false acceptances and detected direct tampering of stock records. Every component attaches evidence to its outputs and protects the data its stakeholders regard as sensitive.

The evidence does not yet support stronger claims. The learning models in C03 and C04 are baselines trained on synthetic data; automated exchanges between components are specified but not verified end to end; and the solution has not been evaluated with farmers, millers or PMB staff in the field. The project therefore demonstrates that the integrated ecosystem is technically feasible, coherent and responsive, but not yet that it improves farmer incomes, market fairness or food security in practice.

8.2 Principal integrated contribution

The contribution of the team goes beyond the four individual components in three ways.

- **A supply-chain-wide decision architecture.** The project shows how farm, price, trade and storage decisions can be supported by independent AI services

that share users, conventions and identifiers, so that each stage's output can inform the next without creating mandatory dependencies or a single point of failure.

- **A consistent trust model for AI in agriculture.** Across very different techniques, the team applied one principle: every output must carry evidence that makes it explainable (SHAP evidence, rule-based scores) or verifiable (confidence gates, validated negotiation, zero-knowledge proofs, ledger anchoring), and estimates must never be presented as guarantees.
- **Privacy scoped to real stakeholder concerns.** Rather than claiming general privacy, each component protects the specific information its users are reluctant to disclose (reservation prices, warehouse capacities, personal data) and states the limits of that protection precisely.

The team also learned that integration evidence must include the meaning of stored data, not only the shape of interfaces, a lesson drawn from defects that only a data reconciliation audit could find.

8.3 Future directions

The following directions are prioritised by how much each would strengthen the evidence for the integrated solution.

- **Real data and field validation (highest priority).** Retrain and evaluate the C02, C03 and C04 models on real yield, transaction and PMB warehouse data with documented, stratified splits; evaluate C01 forecasts against actual subsequent prices; and complete the designed user studies with farmers, millers and PMB staff, measuring usability, comprehension of explanations and trust.
- **Verified end-to-end integration.** Implement and test the cross-component exchanges of Table 4.3, adopt a shared district and variety code list, unify authentication into a single sign-on across components, and add a scripted end-to-end scenario and data reconciliation checks to an automated integration test suite.
- **Strengthened trust mechanisms.** Introduce a transactional outbox for C04 ledger anchoring; bind capacity proofs to warehouse identity in the circuit and

move proof generation to the client; move C03 towards genuinely distributed federated learning with secure aggregation; and protect negotiation limits cryptographically.

- **Broader agronomic and market coverage.** Add NPK sensing and pest detection to C02, more districts and causal drivers to C01, and quality, moisture and transport cost to C03 matching.
- **Deployment and scale.** Move from the two-organisation test network to a multi-orderer Raft configuration with PMB regional offices as organisations, characterise performance with established benchmarking methods, and pilot the ecosystem in one district in partnership with the PMB and a farmer organisation before national scaling.

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APPENDICES

[Guidance - delete before submission: Include only material needed to verify the integrated solution, such as interface specifications, contribution evidence, integrated test matrices, approvals, or extended results. Label each appendix and cite it from the text.]

Appendix A Integrated interface specification

This appendix records the interface conventions and the principal endpoints of each component. Field names are given as implemented in the individual components; component-internal endpoints are documented in the individual theses.

A.1 Common conventions

- Transport: HTTPS for deployed services; JSON request and response bodies, except multipart form data for image upload.
- Authentication: JWT bearer tokens (C03, C04; C04 uses short-lived access tokens with rotating refresh tokens); Firebase authentication project shared with C02 for field-to-user linking.
- Validation: request schemas validated at the boundary (Pydantic in Python services, schema parsing and request validation in Node.js services); invalid requests return client errors (for example HTTP 422 in FastAPI services) before any model, database or ledger is accessed.
- Authorisation: role derived from the authenticated token on the server; forbidden operations return HTTP 403.
- Shared identifiers: district name, paddy type or variety, and user role, using agreed reference values.
- Errors: structured JSON error message suitable for display in English or Sinhala by the client.

A.2 Component endpoints

Table A.1: C01 price forecasting endpoints

Endpoint	Principal inputs	Principal outputs
`POST /predict`	district, date	predicted average paddy price
`POST /forecast`	district, start date, weeks (minimum 1)	weekly predictions with dates
`POST /explanation/evidence`	district, date	predicted and previous price, trend, confidence, risk, outlook, recommendations, SHAP feature contributions
`POST /explanation/llm`	as above, language preference	English and simplified Sinhala explanation; fallback message if the LLM is unavailable

Table A.2: C02 digital farming endpoints

Endpoint	Principal inputs	Principal outputs
`POST /predict_disease`	multipart leaf image (resized to 224 x 224 x 3)	disease class and confidence; "Unknown Type" when maximum probability is below 0.85
`POST /predict_yield_production`	District, Season, Zone, Latitude, Longitude, field_id	yield in kg/ha; environmental values used (sensor or weather fallback); drought or irrigation advisories
Variety recommendation	district, zone, season, salinity, iron toxicity	variety code with growth duration and flood tolerance

Table A.3: C03 marketplace interfaces

Interface	Principal inputs	Principal outputs
Authentication (Express, port 5000)	role-based registration and login	JWT for farmer or miller
Harvest and demand listings	paddy type, quantity, district, price, status	stored listings; status updates (active, sold, fulfilled)
Matching and match selection	listing identifier	top candidates with score out of 100 and category; selection record (unique per pair)
Price reference	district, paddy type, season, quantity	indicative price from the FedAvg-style model
Market assistant	question in English, Sinhala or Singlish	answer grounded in retrieved knowledge-base records
Negotiation (FastAPI, port 8000)	approved match; constraints held server side	round-by-round actions, outcome and agreed price

Table A.4: C04 warehouse coordination route modules

Route module	Principal inputs	Principal outputs
Authentication and users	credentials; user administration by administrators	access and refresh tokens; user records
Warehouses and stock events	warehouse registration with coordinates; stock events with documents	warehouse summaries; stored events with SHA-256 digest and ledger reference
Disasters and redistribution	disaster declaration; redistribution order	ranked candidates by composite score; atomic outflow and inflow legs
Zero-knowledge proofs	warehouse, disaster, threshold	proof verification result (duplicates rejected with conflict)
GNN scoring	current warehouse graph	reliability score per warehouse
Blockchain audit and documents	event or disaster identifier	integrity report (database, recomputation, ledger) and chronological audit timeline

Appendix B Team contribution declaration

We, the undersigned members of research group R26-SE-007, confirm that the summary of individual and shared contributions in Section 4.5 (Table 4.6) is an

accurate and agreed record of our work on the Digital Goviya project and on this common report.

Shared contributions. All four members contributed to the definition of the common problem and objectives, the agreement of the shared technology and interface conventions, the integration of their components with the Digital Goviya client and gateway, and the preparation and review of this common report.